dark planet interposes itself between the star and the observer on Earth; or by sensing a faint wobble in the motion of the star as it's tugged first one way and then another by an otherwise invisible orbiting companion. Spaceborne techniques will be much more sensitive. A Jovian planet going around a nearby star is about a billion times fainter than its sun; nevertheless, a new generation of ground-based telescopes that can compensate for the twinkling in the Earth's atmosphere may soon be able to detect such planets in only a few hours' observing time. A terrestrial planet of a neighboring star is a hundred times fainter still; but it now seems that comparatively inexpensive spacecraft, above the Earth's atmosphere, might be able to detect other Earths. None of these searches has succeeded yet, but we are clearly on the verge of being able to detect at least Jupiter-sized planets around the nearest stars —if there are any to be found.

A most important and serendipitous recent discovery is of a bona fide planetary system around an unlikely star, some 1,300 light-years away, found by a most unexpected technique: The pulsar designated B1257+12 is a rapidly rotating neutron star, an unbelievably dense sun, the remnant of a massive star that suffered a supernova explosion. It spins, at a rate measured to impressive precision, once every 0.006218531938818 7 seconds, This pulsar is pushing 10,000 rpm.

Charged particles trapped in its intense magnetic field generate radio waves that are cast across the Earth, about 160 flickers a second. Small but discernible changes in the flash rate were tentatively interpreted by Alexander Wolszczan, now at Pennsylvania State University, in 1991—as a tiny reflex motion of the pulsar in response to the presence of planets. In 1994 the predicted mutual gravitational interactions of these planets were confirmed by Wolszczan from a study of timing residuals at the microsecond level over the intervening years. The evidence that these are truly new planets and not starquakes on the neutron star surface (or something) is now overwhelming—or, as Wolszczan put it, "irrefutable"; a new solar system is "unambiguously identified." Unlike all the other techniques, the pulsar timing method makes